

Electronics Viability Research

A component-by-component check of every part on the flight computer list, based on real datasheet specs and documented issues — not just “this is a common part.” Goal: know exactly what could go wrong before ordering anything.

1. Microcontroller — Arduino Nano (Clone, CH340)

Spec	Value
Logic level	5 volts (ATmega328P is not 3.3-volt tolerant on its I/O pins)
USB-serial chip	CH340G
Input voltage (Vin)	7–12 volts recommended
Onboard regulator	AMS1117-5.0 linear regulator

Viability: Yes, with two known gotchas to plan around.

This is the standard, well-supported choice it's reputed to be — the concern isn't whether it works, it's two specific integration details that trip people up.

- **5-volt logic, not 3.3-volt tolerant:** connecting a 5-volt output pin directly to a 3.3-volt sensor input can damage the sensor. This mostly isn't a problem here because breakout boards for the BMP388, GPS, and SD module used in this build already include their own onboard voltage regulation and level shifting — the raw bare chips would be the risk, not these breakout modules.
- **CH340 driver issues on Windows:** WCH's official driver (version 3.8 and newer) intentionally blocks unauthorized clone silicon, which can show up as a “Code 10, this device cannot start” error in Device Manager. Fix is well-documented: uninstall the current driver and install the older 3.4.2014.8 version, which doesn't have the clone-blocking check. Not a hardware defect — a five-minute software fix.

2. Altitude — BMP388 Barometric Sensor

Spec	Value
Pressure range	300–1,250 hPa (covers roughly sea level up to about 9,000 meters / 29,500 feet)
Sampling rate	Up to 200 Hz
Relative accuracy	±8 Pa, approximately ±0.5 meters of altitude
Interface	I2C or SPI
Mechanical shock rating	12,000 g absolute maximum (per datasheet)

Viability: Strong yes.

This isn't a borderline case. The pressure range alone covers more than 8 times your 3,500 foot (1,067 meter) target altitude, so you're nowhere near its ceiling. It's also already the primary altimeter chip in several open-source flight controller projects (Betaflight, ArduPilot), and hobbyists have specifically flight-tested it on model rockets and reported it working well, with the caveat that pressure-based altitude needs local sea-level pressure entered for an accurate absolute reading — relative altitude (how high above the pad) is accurate regardless.

3. Accelerometer/Gyro — MPU6050

Spec	Value
Accelerometer range	Selectable: $\pm 2g$, $\pm 4g$, $\pm 8g$, or $\pm 16g$ (software-configured; defaults to $\pm 2g$)
Gyroscope range	Selectable up to $\pm 2,000^\circ/\text{second}$
Interface	I2C
Current draw	Less than 3.6 mA active, 5 μA idle

Viability: Conditional — needs a specific code setting, and even then has a real ceiling.

This is the one component on the list with a genuine, non-hypothetical risk worth flagging before you buy.

- **The default range is wrong for this application.** Out of the box, the MPU6050 is usually configured for $\pm 2g$ — fine for detecting tilt or footsteps, but a rocket launch will slam it to full-scale (clipped, flat-lined) output almost instantly. This must be explicitly set to $\pm 16g$ in your firmware before flight.
- **$\pm 16g$ is the hardware ceiling — there's no higher setting on this chip.** Peak acceleration for a G-motor rocket depends heavily on total mass, but published OpenRocket/RockSim examples for lightweight rockets on much smaller motors have shown 20–50+ g peaks; a heavier rocket like this one (roughly 350 grams dry, more with a motor) will land considerably lower, plausibly in the 10–20 g range depending on the exact motor chosen — which means the very peak of a boost curve could clip the sensor even at its highest setting.
- **Practical impact is limited, not disqualifying:** a clipped peak reading only affects the single instant of maximum thrust — it does not prevent the barometer from independently tracking real altitude, and it does not prevent the gyroscope (a separate sensor on the same chip, unaffected by this limit) from tracking orientation. This is a known, documented rocketry forum discussion point, not a novel concern.
- **Mitigation if precise peak-G data matters to you:** an ADXL345 ($\pm 16g$, similar cost) offers no real improvement, but an ADXL375 ($\pm 200g$ range) is a direct swap-in used specifically by rocketry hobbyists for this exact reason, at a similar price point. Worth considering if capturing an accurate acceleration curve matters more to you than sticking to the original parts list — otherwise the MPU6050 is fine for its primary job here (orientation/attitude, with altitude coming mainly from the barometer).

4. Position — NEO-6M GPS Module

Spec	Value
Update rate	1 Hz default, configurable up to 5 Hz
Time-to-first-fix	Cold start ~29–32 s, warm start ~23–25 s, hot start <1 s
Operational altitude limit	50,000 meters (164,000 feet) — far above your target
Operational velocity limit	515 m/s (COCOM export-control limit)
Operational acceleration limit	4 g, per manufacturer datasheet
Supply voltage	3.6–5.5 volts (module typically has onboard regulation)

Viability: Yes for its actual job here (recovery), with one caveat during boost.

The altitude and velocity limits you'd read about generically (COCOM export restrictions) are a total non-issue at 3,500 feet and G-motor speeds — you're nowhere close to either ceiling. The real number worth knowing is one that doesn't get mentioned as often:

- **The datasheet also lists a 4g acceleration operating limit** — separate from the altitude/velocity COCOM limit. A rocket in boost phase will very likely exceed 4g, meaning the GPS may lose or fail to compute a valid position fix for the roughly 1–3 seconds the motor is burning.
- **This matters far less than it sounds like it should.** The GPS's actual job in this build is helping you find the rocket after it lands — by the time the parachute has deployed and the rocket is descending, acceleration is back down near 1g (or a bit more from parachute deceleration), well inside the GPS's normal operating range. A brief dropout during the few seconds of boost does not prevent it from reacquiring a fix during the multi-second-to-minute coast and descent phases that follow.
- **Net assessment:** viable as specified for locating the rocket after recovery, not viable as a high-fidelity in-flight trajectory sensor during boost — which was never really its job here anyway (that's the barometer and IMU's role).

5. Storage — MicroSD Card Module (SPI)

Spec	Value
Card logic voltage	3.3 volts (hard requirement — 5 volts will damage the card)
Interface	SPI (4 signal lines: MOSI, MISO, SCK, CS)
Typical current draw	15–30 mA read, up to ~100 mA during writes
Compatible library	Arduino built-in SD library (FAT16/FAT32)

Viability: Yes — but only the correct module variant.

There are two different products sold under “microSD module” and they are not interchangeable with a 5-volt Arduino:

- **“Arduino-ready” modules (what you want):** these include an onboard 3.3-volt regulator plus a 74LVC125A logic-level-shifter chip, specifically so they can be wired directly to a 5-volt board's SPI pins without frying the card. This is the standard, widely-sold version and the one to search for.
- **Bare/raw SD sockets (avoid for this build):** no onboard regulation or level shifting — wiring one of these straight to the Nano's 5-volt logic can permanently damage the microSD card. These are cheaper but require you to build external level-shifting circuitry yourself, which isn't worth the savings here.
- **Buying tip:** check the listing photo for a 6-pin module with a small chip visible near the card slot (the regulator + level shifter) — that's the one you want.
- **Vibration note:** under sustained vibration or a hard landing, a loosely-seated card can lose contact momentarily. Cheap insurance: a small dab of hot glue or a strip of tape across the card slot after final testing, so it can't work itself loose in flight.

6. Radio — 2× RFM95W LoRa Modules (915 MHz)

Spec	Value
Chipset	Semtech SX1276
Range, simple wire antenna	Approximately 2 kilometers line-of-sight
Range, directional antenna + tuning	Up to approximately 20 kilometers
Interface	SPI
Supply voltage	3–5 volts (onboard regulator and level shifter included)
Current draw	~100 mA peak during transmit (+20 dBm), ~30 mA while listening

Viability: Strong yes, with generous margin.

Your target apogee of 3,500 feet is about 1,067 meters — well under 2 kilometers — and horizontal drift for a rocket this size on a G motor is typically a few hundred meters at most under parachute. Even the simplest wire-antenna setup has real margin here; you're unlikely to need a directional antenna at all. The onboard regulator and level shifter mean it's also safe to wire directly to the 5-volt Nano without extra components.

- **Antenna is not optional:** you need to solder a short length of wire (a simple monopole) to the module, or add a uFL/SMA connector for a screw-on antenna. It won't transmit meaningfully without one.

- **Frequency matching:** both radios must be the same band variant (915 MHz to 915 MHz) — they cannot talk to a 433 MHz module or a different radio family (like RFM69). Confirm both listings say 915 MHz specifically when ordering.
- **Sharing the SPI bus with the SD module:** both the radio and the SD card use SPI, which is fine — SPI is designed for multiple devices on one bus, distinguished by separate chip-select (CS) pins. This is a very common, well-supported pattern, not a conflict.

7. Power — 1S LiPo Battery + Voltage Regulator

Spec	Value
Battery capacity	500–900 mAh, single-cell (1S) LiPo
Nominal voltage	3.7 volts (LiPo), needs regulation up to 5 volts for the Nano and most modules
Estimated peak combined current draw	~150–200 mA (radio TX ~100 mA + GPS ~45 mA + sensors/MCU overhead)

Viability: Yes, comfortable margin for a short flight.

A full flight from arm to recovery is measured in minutes, not hours. Even at a generous continuous 200 mA draw, a 500 mAh battery provides well over an hour of runtime — many times more than needed. The real requirements here are basic but non-negotiable:

- **Buy a LiPo with a built-in protection PCB** (nearly all hobbyist 1S LiPos sold with a standard JST-PH connector include this) — it protects against over-discharge, which is easy to do accidentally on a project you might leave powered on between bench tests.
- **Match the regulator to what your specific modules need.** If everything on the sled is 5-volt logic (Nano, most breakout boards), a single buck/boost regulator stepping the LiPo's 3.7–4.2 volts up to a steady 5 volts is sufficient — confirm this against the actual modules you buy, since some GPS or radio modules are more efficient run directly at 3.3 volts.

Bottom line: every component on the original list is genuinely viable for this build. The MPU6050's accelerometer range and the GPS's boost-phase acceleration limit are worth knowing about going in, but neither is a reason to abandon the part — they're limitations to design around (correct firmware configuration, and realistic expectations about what each sensor is actually responsible for measuring), not defects that will strand the project.

Generated as a pre-purchase viability check for the 3,500 ft rocket project's electronics list.